

STAT1201 Analysis of Scientific Data – 202p
Sample Examination - Solutions

Question 1

- a) $E(X) = 0.2 \times 0 + 0.4 \times 1 + 0.3 \times 2 + 0.1 \times 3 = 1.3$
 $\text{var}(X) = 0.2 \times (0 - 1.3)^2 + \dots + 0.1 \times (3 - 1.3)^2 = 0.81$, so
 $\text{sd}(X) = \sqrt{0.81} = 0.9$
- b) $E(Y) = 40 E(X) = 52$
 $\text{sd}(Y) = \sqrt{40} \text{sd}(X) = 5.69$
- c) $P(Y \geq 60) = P\left(Z \geq \frac{60-52}{5.69}\right) = P(Z \geq 1.405) = 0.080$, an 8% chance of at least 60.
- d) The .5 quantile is 0 so we want $\frac{60-1.3n}{.9\sqrt{n}} = 0$, giving $n = \frac{60}{1.3} = 46.15$. Thus, they would need at least 47 mother plants to have a 50% chance of 60 cuttings.

Question 2

a) Observational. The BD and UD status of participants was not manipulated.

b) The completed table is as follows:

Source	df	SS	MS	F	p
Group	2	482.3	241.15	9.32	.0002722
Residuals	66	1706.8	25.86		
Total	68	2189.1			

c) $R^2 = \frac{482.3}{2189.1} = .22$, so the groups explain 22% of the variability in PONS test scores.

d) The BD vs Control and UD vs Control differences are significant but not the BD vs UD difference. Overall it seems BD and UD subjects have significantly lower PONS scores compared to the controls but their means are similar to each other.

e) Expected values are

Group	Age	
	Under 55	Over 55
BD	7 11.3	16 11.7
UD	18 11.3	5 11.7
Control	9 11.3	14 11.7

$$\chi^2_2 = \frac{(7 - 11.3)^2}{11.3} + \frac{(16 - 11.7)^2}{11.7} + \dots + \frac{(14 - 11.7)^2}{11.7} = 11.9$$

p-value is $P(\chi^2_2 \geq 11.9) = 1 - .9974 = .0026$, giving strong evidence of a difference in age distribution between the three groups.

Question 3

- a) Let μ_1, μ_2 be the mean birth weights for non-smoking and smoking mothers, respectively.

Test $H_0: \mu_1 = \mu_2$ vs $H_1: \mu_1 > \mu_2$

$$t = \frac{(3292 - 2982) - 0}{\sqrt{\frac{505^2}{40} + \frac{694^2}{10}}} = \frac{310}{233.5} = 1.33, \text{ so } p\text{-value is } P(T_9 \geq 1.33) = .108$$

Inconclusive evidence that the mean birth weight is lower for smokers.

- b) $2 \times 87.68 = 175.36$ g.
- c) Let β_1 be the slope of the relationship between birth weight and weeks of gestation.
Test $H_0: \beta_1 = 0$ vs $H_1: \beta_1 > 0$ using

$$t_{48} = \frac{87.68 - 0}{22.26} = 3.94$$

p -value is $P(T_{48} \geq 3.94) = .00013$, very strong evidence of a positive association between birth weight and weeks of gestation.

- d) For a given number of weeks of gestation, the mean birth weight for smoking mothers is 357.43 g lower than for non-smoking mothers.
- e) There is strong evidence that birth weight is associated with weeks of gestation, after taking smoking status into account, and there is moderate evidence of a difference in birth weights between smoking and non-smoking mothers, after taking weeks of gestation into account.
- f) This is a Normal Quantile Plot of the residuals that shows the Normality assumption is questionable for this model since the points seem to deviate from the diagonal line.

Question 4

a) Test $H_0: \mu = 0$ vs $H_1: \mu > 0$

b) Test using

$$t_9 = \frac{5.15 - 0}{5.54/\sqrt{10}} = 2.94$$

So p-value is $P(T_9 \geq 2.94) = 0.0082$, giving strong evidence that there is a mean increase in pulse rate with caffeine-free cola.

c) 95% CI is $5.15 \pm 2.262 \times \frac{5.54}{\sqrt{10}} = 5.15 \pm 3.96$ bpm, or (1.19, 9.11) bpm.

d) 9 out of the 10 differences are positive, giving p-value for sign test $P(X \geq 9) = .0107$, so sign test gives strong evidence that caffeine-free cola tends to increase pulse rate.

e) The ranked observations are

4.4	10.1	7.2	-8.6	4.5	3.6	5.4	6.5	6.0	12.4
2	9	7	8	3	1	4	6	5	10

Counting positives gives

$$S = 2 + 9 + 7 + 3 + 1 + 4 + 6 + 5 + 10 = 47$$

with

$$E(S) = \frac{10 \times 11}{4} = 27.5$$

and

$$sd(S) = \sqrt{\frac{10 \times 11 \times 21}{24}} = 9.81$$

p-value is $P\left(Z \geq \frac{47-27.5}{9.81}\right) = P(Z \geq 1.99) = .023$, moderate evidence to suggest that caffeine-free cola tends to increase pulse rate.

The evidence is weaker since the signed-rank test takes into account that the one negative value was relatively large.